

PROJECT			JOB NO.	
MAIN CONTRACTOR				
SUB-CONTRACTOR		: WINNER ENGINEERING PTE LTD		
TRADE		: ACMV		
AHU/FCU PRE -CON CHECK LIST				
S/N	Description	Existing Information/Status	Measurement Reading	Remarks
1	IDentification			
1.1	Identification Number			
1.2	Location			
1.3	Serial Number			
1.4	Model Number			
1.5	Fan Airflow Capacity (m ³ /h or m/s)			
1.6	Cooling Capacity (kW / RT)			
1.7	Motor Power (kW)			
2	Physical Inspection			
2.1	Filter Condition & Size			
2.2	Ceiling Access Panel Clearance & Size			
2.3	General Corrosion & Structural Damage			
2.4	Drive Belt Condition & Tension			
2.5	Fan Blade & Impeller Inspection			
2.6	Motor & Fan Drive Assembly			
2.7	Belt Specification & Model			
2.8	Differential Pressure Switch (Filter Status)			
2.9	Motorized Damper Operation			
2.10	Non-Return Damper (NRD) Inspection			
2.1	Duct Smoke Detectors Check			
2.12	Fire Dampers & Fusible Link Check			
2.1	CO ₂ / Gas Sensors: Inspect sensor condition, mounting, and calibration status.			
2.14	Main Duct Dimensions (Supply / Return)			
2.2	Main Return Duct Volume Control Damper (VCD)			
2.16	Condensate Drainage & Pitch			
2.2	Drain Pipe Outlet & 'U' Trap Integrity			
2.18	Condensate Drain Pan Cleanliness			
2.2	Duct Insulation Condition (Damage / Condensation)			
2.20	Chilled Water Piping Insulation (Damage / Sweating)			
2.2	Chilled Water Thermometer Condition			
2.22	Chilled Water Gate / Isolation Valves			
2.2	Chilled Water Pressure Gauge Condition			
2.24	Equipment Casing Damage & Condensation			
2.3	Noise Level Measurement (dB)			
2.26	Vibration Isolators & Springs			
2.3	UV Germicidal Lamp Condition			
3	Air Side			
3.1	Total Airflow Rate (m ³ /h or m/s)			
3.2	VSD Model & Fan Speed Settings (Hz)			
3.3	Supply Air Temperature (°C)			
3.4	Return Air Temperature (°C)			
3.5	Room Temperature Setpoint (°C)			
3.6	On-Coil Air Temperature (°C)			
3.7	Off-Coil Air Temperature (°C)			
3.8	Relative Humidity (%RH)			
3.9	Duct Static Pressure Set Point (DSP) Measured Duct Static Pressure (DSP)			
3.1	Duct & System Air Leakage Check			

4	BMS System										
4.1	BMS Control System Integration Check										
4.2	Thermostats & Sensor Calibration Verification										
4.3	Local Control Panel (LCP): Verify Switch Operation										
4.4	VSD Speed Signal Modulation (0-10V / 4-20mA)										
5	Water Side										
5.1	Cooling Coil Condition & Fin Inspection										
5.2	Coil Corrosion & Fouling Inspection										
5.3	Chilled Water Pressure Gauge Readings (Supply/Return)										
5.4	Chilled Water Supply & Return Temperatures (CHWST/CHWRT)										
5.5	Chilled Water Control Valve Actuator Operation										
5.6	Valve 100% Open - On-Coil Water/Air Temp (°C)										
5.7	Valve 100% Open - Off-Coil Water/Air Temp (°C)										
5.8	Valve 100% Closed - On-Coil Water/Air Temp (°C)										
5.9	Valve 100% Closed - Off-Coil Water/Air Temp (°C)										
5.10	Chilled Water Pipe Diameter (mm / inch)										
5.11	Water Leak Detection Sensor Check										
6	Electrical Connections										
6.1	Electrical Visual Inspection & Phase Verification										
6.2	Wiring Integrity & Earthing Check										
6.3	Panel Single-Line Diagram (SLD) Schematic Display										
6.4	Motor Starting Current (A)										
6.5	Motor Running Current (A)										
6.6	Incoming Isolator Switch Operation										
7	Maintenance Records										
Note : _____ _____											
<table border="1"> <tr> <td rowspan="3">Legends:</td> <td>√</td> <td>CHECKED</td> </tr> <tr> <td>NA</td> <td>NOT APPLICABLE</td> </tr> <tr> <td>×</td> <td>NOT ABLE TO CHECK</td> </tr> </table>					Legends:	√	CHECKED	NA	NOT APPLICABLE	×	NOT ABLE TO CHECK
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Installed & Tested By
Sub- Contractor

(Name & Signature)

Date & Time

Inspected & Checked BY
Main Contractor

(Name & Signature)

Date & Time

Witnessed & Accepted By
Consultant / RTO

(Name & Signature)

Date & Time